DATABASE DESIGN & ER MODEL

Detailed relational schema, constraints and data ownership

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

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1. Purpose

This document defines the persistent data model for the MCI assistive application. It converts the approved requirements into a concrete database structure that can be implemented before feature coding begins. The design separates deterministic application data from semantic memory data.

2. Design Principles

- Use a relational database for users, patients, schedules, medications, appointments, reminders, alerts and configuration.
- Use a vector index/store for conversation/object/person memory embeddings; keep a relational metadata record for every vectorized item.
- Use foreign keys to enforce patient ownership and caregiver relationships.
- Use timestamps consistently for observations, conversations, reminders, locations and alerts.
- Do not store raw biometric media unless required; store controlled recognition representations and metadata.
- Treat last-seen object information as an observation record, not as a guaranteed current state.

3. ER Diagram

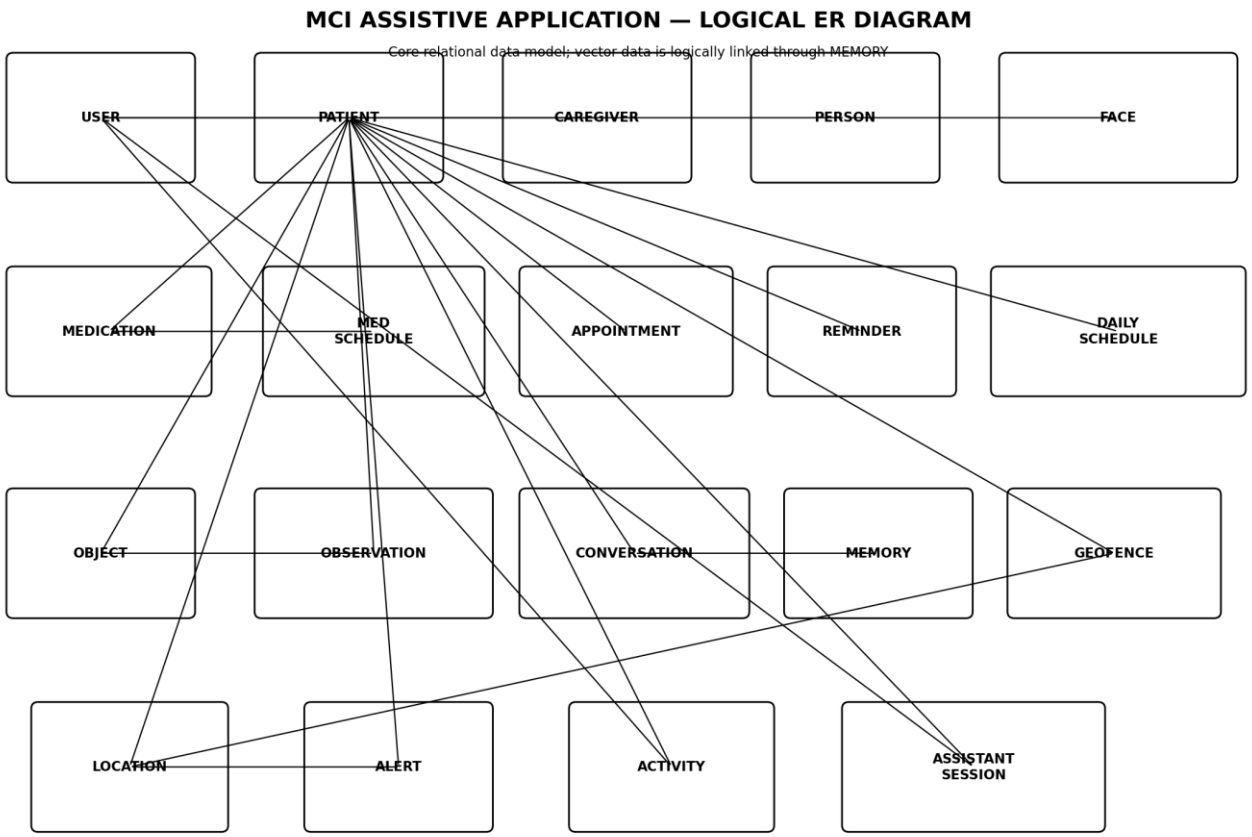


Figure 1 — Logical ER model. The diagram shows the principal relationships; implementation may normalize or split tables further.

4. Table Catalogue

Table	Purpose	Primary Key
users	Authentication and common account information	id
patients	Patient-specific profile	id
caregivers	Caregiver-specific profile	id
caregiver_patient	Authorized caregiver-to-patient relationship	id
people	Familiar-person registry	id
face_embeddings	Recognition representations	id
objects	Tracked object definitions	id
object_observations	Object detection/last-seen observations	id
conversations	Conversation/session metadata	id
conversation_messages	Individual messages/transcript segments	id
memories	Vectorized searchable memory metadata	id
medications	Medication master records	id
medication_schedules	Medication timing/recurrence	id
appointments	Calendar appointments	id
daily_schedules	Routine activities	id
reminders	Reminder definitions/events	id
geofences	Configured safe zones	id
locations	Location observations	id
alerts	Caregiver safety/important notifications	id
activity_logs	Auditable system/user events	id
assistant_sessions	AI assistant sessions	id

5. Detailed Schema

users

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Unique user identifier
name	VARCHAR(120)	NOT NULL	Display name
email	VARCHAR(255)	UNIQUE	Login/contact identifier
phone	VARCHAR(30)	NULL	Optional phone
password_hash	TEXT	NOT NULL	Hashed credential
role	ENUM	NOT NULL	PATIENT / CAREGIVER / ADMIN if required
is_active	BOOLEAN	NOT NULL	Account status
created_at	TIMESTAMP	NOT NULL	Creation time
updated_at	TIMESTAMP	NOT NULL	Last modification

patients

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Patient profile identifier
user_id	FK users.id	UNIQUE, NOT NULL	Patient login account
date_of_birth	DATE	NULL	Optional profile field
preferred_language	VARCHAR(20)	NULL	UI/voice preference
timezone	VARCHAR(64)	NOT NULL	Reminder timezone
notes	TEXT	NULL	Non-clinical caregiver notes
created_at	TIMESTAMP	NOT NULL	Creation time
updated_at	TIMESTAMP	NOT NULL	Last modification

caregivers

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Caregiver profile identifier
user_id	FK users.id	UNIQUE, NOT NULL	Caregiver login account
created_at	TIMESTAMP	NOT NULL	Creation time
updated_at	TIMESTAMP	NOT NULL	Last modification

caregiver_patient

Column	Type	Constraint	Description

id	UUID / BIGINT	PK	Relationship identifier
caregiver_id	FK caregivers.id	NOT NULL	Authorized caregiver
patient_id	FK patients.id	NOT NULL	Managed patient
permission_level	VARCHAR(30)	NOT NULL	Configured access level
is_active	BOOLEAN	NOT NULL	Relationship status
created_at	TIMESTAMP	NOT NULL	Creation time

people

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Familiar person
patient_id	FK patients.id	NOT NULL	Patient who knows this person
name	VARCHAR(120)	NOT NULL	Person name
relationship	VARCHAR(80)	NULL	Relationship such as daughter/friend
notes	TEXT	NULL	Optional contextual notes
is_active	BOOLEAN	NOT NULL	Recognition status
created_at	TIMESTAMP	NOT NULL	Creation time

face_embeddings

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Embedding record
person_id	FK people.id	NOT NULL	Registered person
embedding_ref	TEXT	NOT NULL	Reference to vector storage
model_name	VARCHAR(120)	NOT NULL	Embedding/recognition model
model_version	VARCHAR(60)	NULL	Model version
quality_score	DECIMAL	NULL	Quality metric if available
created_at	TIMESTAMP	NOT NULL	Creation time

objects

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Tracked object definition
patient_id	FK patients.id	NOT NULL	Owner/context
name	VARCHAR(100)	NOT NULL	Object label
category	VARCHAR(80)	NULL	Object category
is_active	BOOLEAN	NOT NULL	Tracking status
created_at	TIMESTAMP	NOT NULL	Creation time

object_observations

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Observation
object_id	FK objects.id	NOT NULL	Object
patient_id	FK patients.id	NOT NULL	Patient context
observed_at	TIMESTAMP	NOT NULL	Detection time
location_label	VARCHAR(160)	NULL	Room/scene label
latitude	DECIMAL	NULL	Optional location
longitude	DECIMAL	NULL	Optional location
confidence	DECIMAL	NULL	Detection confidence
image_ref	TEXT	NULL	Controlled reference if an image is retained

conversations

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Conversation session
patient_id	FK patients.id	NOT NULL	Patient
recognized_person_id	FK people.id	NULL	Person context if known
started_at	TIMESTAMP	NOT NULL	Start
ended_at	TIMESTAMP	NULL	End
source	VARCHAR(40)	NOT NULL	VOICE / TEXT / OTHER
retention_status	VARCHAR(30)	NOT NULL	Retention policy state

conversation_messages

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Message
conversation_id	FK conversations.id	NOT NULL	Conversation
speaker_type	VARCHAR(30)	NOT NULL	PATIENT / PERSON / SYSTEM
content	TEXT	NOT NULL	Transcript/text

created_at	TIMESTAMP	NOT NULL	Message time

memories

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Searchable memory
patient_id	FK patients.id	NOT NULL	Patient
source_type	VARCHAR(40)	NOT NULL	CONVERSATION / OBJECT / EVENT / OTHER
source_id	UUID / BIGINT	NOT NULL	Source record
content_ref	TEXT	NULL	Controlled reference/text payload
vector_ref	TEXT	NOT NULL	Vector index identifier
recognized_person_id	FK people.id	NULL	Person context
event_time	TIMESTAMP	NULL	Original event time
created_at	TIMESTAMP	NOT NULL	Index time

medications

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Medication
patient_id	FK patients.id	NOT NULL	Patient
name	VARCHAR(160)	NOT NULL	Medication name
dosage_text	VARCHAR(160)	NULL	Configured dosage/instruction
instructions	TEXT	NULL	Caregiver-provided instructions
is_active	BOOLEAN	NOT NULL	Active status
created_at	TIMESTAMP	NOT NULL	Creation time
updated_at	TIMESTAMP	NOT NULL	Last modification

medication_schedules

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Schedule
medication_id	FK medications.id	NOT NULL	Medication
time_of_day	TIME	NOT NULL	Reminder time
recurrence_rule	TEXT	NOT NULL	Daily/weekly/custom recurrence
start_date	DATE	NULL	Start
end_date	DATE	NULL	End
is_active	BOOLEAN	NOT NULL	Active

appointments

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Appointment
patient_id	FK patients.id	NOT NULL	Patient
title	VARCHAR(180)	NOT NULL	Appointment title
start_at	TIMESTAMP	NOT NULL	Start
end_at	TIMESTAMP	NULL	End
location	VARCHAR(255)	NULL	Location
notes	TEXT	NULL	Notes
created_by	FK users.id	NOT NULL	Creator
created_at	TIMESTAMP	NOT NULL	Creation time

daily_schedules

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Routine item
patient_id	FK patients.id	NOT NULL	Patient
title	VARCHAR(180)	NOT NULL	Activity
scheduled_time	TIME	NOT NULL	Time
recurrence_rule	TEXT	NOT NULL	Recurrence
priority	VARCHAR(20)	NULL	Priority
is_active	BOOLEAN	NOT NULL	Active

reminders

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Reminder
patient_id	FK patients.id	NOT NULL	Patient
type	VARCHAR(40)	NOT NULL	MEDICATION / APPOINTMENT / ROUTINE / OTHER
reference_id	UUID / BIGINT	NULL	Related entity
scheduled_at	TIMESTAMP	NOT NULL	Occurrence

message	VARCHAR(255)	NOT NULL	Reminder text
status	VARCHAR(30)	NOT NULL	PENDING / SENT / ACKNOWLEDGED / MISSED

geofences

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Safe zone
patient_id	FK patients.id	NOT NULL	Patient
name	VARCHAR(120)	NOT NULL	Zone name
latitude	DECIMAL	NOT NULL	Center latitude
longitude	DECIMAL	NOT NULL	Center longitude
radius_m	DECIMAL	NOT NULL	Radius in meters
is_active	BOOLEAN	NOT NULL	Active
created_by	FK users.id	NOT NULL	Creator
created_at	TIMESTAMP	NOT NULL	Creation time

locations

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Location event
patient_id	FK patients.id	NOT NULL	Patient
latitude	DECIMAL	NOT NULL	Latitude
longitude	DECIMAL	NOT NULL	Longitude
accuracy_m	DECIMAL	NULL	Location accuracy
recorded_at	TIMESTAMP	NOT NULL	Time
source	VARCHAR(40)	NOT NULL	GPS / network / test

alerts

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Alert
patient_id	FK patients.id	NOT NULL	Patient
type	VARCHAR(50)	NOT NULL	GEOFENCE / MEDICATION / SYSTEM / OTHER
severity	VARCHAR(20)	NOT NULL	INFO / WARNING / HIGH
message	TEXT	NOT NULL	Alert content
location_id	FK locations.id	NULL	Related location
created_at	TIMESTAMP	NOT NULL	Creation
acknowledged_at	TIMESTAMP	NULL	Acknowledgement
acknowledged_by	FK users.id	NULL	Acknowledging user

activity_logs

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Log
patient_id	FK patients.id	NULL	Patient context
actor_user_id	FK users.id	NULL	Actor
event_type	VARCHAR(80)	NOT NULL	Event type
event_data	JSON	NULL	Structured metadata
created_at	TIMESTAMP	NOT NULL	Event time

assistant_sessions

Column	Type	Constraint	Description
id	UUID / BIGINT	PK	Assistant session
patient_id	FK patients.id	NOT NULL	Patient
started_at	TIMESTAMP	NOT NULL	Start
ended_at	TIMESTAMP	NULL	End
mode	VARCHAR(30)	NOT NULL	TEXT / VOICE
context_person_id	FK people.id	NULL	Recognized person context

6. Key Relationships and Cardinalities

Relationship	Cardinality	Rule
User → Patient	1 : 0..1	A patient profile belongs to one user account.
User → Caregiver	1 : 0..1	A caregiver profile belongs to one user account.

Caregiver ↔ Patient	M : N	Use caregiver_patient for explicit authorization.
Patient → Person	1 : N	Patient can have multiple familiar people.
Person → FaceEmbedding	1 : N	Multiple recognition representations may exist.
Patient → Object	1 : N	Patient context contains tracked objects.
Object → Observation	1 : N	Each object can have many observations.
Patient → Conversation	1 : N	Patient has multiple conversations.
Conversation → Message	1 : N	Conversation contains messages.
Conversation/Event → Memory	1 : N	Memory records are vector-search metadata.
Patient → Medication	1 : N	Patient can have multiple medications.
Medication → MedicationSchedule	1 : N	Medication can have multiple schedules.
Patient → Appointment	1 : N	Appointments belong to patient.
Patient → DailySchedule	1 : N	Routine activities belong to patient.
Patient → Reminder	1 : N	Generated reminder occurrences belong to patient.
Patient → Geofence	1 : N	Multiple safe zones may be configured.
Patient → Location	1 : N	Location events are time-series records.
Patient → Alert	1 : N	Alerts belong to patient.

7. Indexing Strategy

- users.email — unique index.
- caregiver_patient(caregiver_id, patient_id) — composite unique/index.
- object_observations(object_id, observed_at DESC) — fast last-seen retrieval.
- conversations(patient_id, started_at DESC).
- memories(patient_id, event_time DESC).
- memories(patient_id, recognized_person_id) — person-aware retrieval metadata.
- medication_schedules(medication_id, time_of_day).
- reminders(patient_id, scheduled_at, status).
- locations(patient_id, recorded_at DESC).
- alerts(patient_id, created_at DESC, acknowledged_at).
- activity_logs(patient_id, created_at DESC).

8. Data Retention and Privacy Rules

- Raw camera frames should not be persisted by default unless required for a defined feature.
- Raw audio should not be persisted by default; transcripts should be retained only according to the project retention policy.
- Face embeddings require controlled access and should be deletable when a familiar-person profile is revoked.
- Location history should have a defined retention window; the project should avoid indefinite tracking by default.
- Conversation memory should have an explicit retention status and deletion path.
- Caregiver access must always be checked against caregiver_patient authorization.

9. Implementation Notes

The schema is intentionally database-agnostic at the logical level. A Django implementation can map these entities to Django models and migrations. UUIDs are recommended where practical. JSON fields may be used for flexible metadata, but core business fields should remain typed relational columns.

10. Database Completion Checklist

- All models created.
- Foreign-key relationships validated.
- Unique constraints added.
- Indexes added for common retrieval paths.
- Migration generated and applied.
- Seed/test data created.
- Authorization filters tested.
- Delete/retention behavior tested.
- Vector-store references tested against relational memory records.